HARSH KUMAR

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PROFESSIONAL SUMMARY

Junior AI & Data Engineer with hands-on experience building Generative AI solutions, LLM-powered pipelines, and real-time data dashboards for the banking and finance sector. Proficient in Python, FastAPI, RAG architecture, and SQL-based BI tools (Power BI, SSRS, Plotly Dash). Automated manual workflows, improved reporting accuracy by 40%, and delivered Text-to-SQL systems with 95%+ precision. Focused on applying AI to solve real-world business problems and drive data-informed decisions.

SKILLS

- Generative AI & LLM Orchestration: Retrieval-Augmented Generation (RAG), LLM Integration (OpenAI, LLaMA), Prompt Engineering, Few-Shot Prompting, Text-to-SQL Pipelines
- Backend & API Development: FastAPI, RESTful API Design, Metadata Discovery Engines, Database Schema Mapping
- Database Management & BI: SQL Server (SSMS), PostgreSQL, SSRS (RDL Report Development), Stored Procedure Automation, T-SQL, SSIS ETL Pipelines
- Data Visualization & Dashboards: Plotly Dash, Power BI (DAX, Data Modeling), Dash Bootstrap Components, Custom CSS
- Languages & Tools: Python (Pandas, NumPy, SQLAlchemy, psycopg2), GitHub, Visual Studio 2019

EXPERIENCE

Natural Consult Private Limited May 2025 – Present

Python Backend & Data Engineer (AI/LLM Systems & Banking Analytics) *Jaipur, Rajasthan*

- Led development of a real-time MIS dashboard for executive leadership, consolidating core banking KPIs (NPA, CASA, CD Ratio) across Head Office and branch levels — enabling faster, data-driven decisions and replacing manual weekly reporting.
- Managed end-to-end SSIS ETL pipelines across heterogeneous banking servers, achieving 100% data consistency and ensuring clean, audit-ready datasets for regulatory submissions.
- Delivered automated SSRS audit reports (ALM, OSS) that cut manual data extraction effort by 40%, directly reducing compliance team workload ahead of regulatory deadlines.
- Deployed a Text-to-SQL AI system into production that empowered non-technical banking staff to independently query live financial data — reducing dependence on the data team for ad hoc report requests.
- Iterated on LLM prompt engineering and metadata retrieval strategies based on user feedback, improving query accuracy from initial baseline to 95%+ precision in real-world banking scenarios.

NullClass Pvt. Ltd July 2024 – August 2024

Data Analyst Trainee *Remote*

- Analyzed social media metrics using Python to uncover viral trends, optimizing audience engagement workflows by 25% through data-driven insights.
- Developed interactive Power BI visualizations (dual-axis and drill-down charts) to monitor tweet engagement rates, enabling real-time tracking and performance benchmarking.
- Pinpointed peak engagement periods through comparative analysis of weekday vs. weekend posting data, driving a 20% increase in strategic engagement efficiency.

PROJECTS

BankBot — Conversational Banking Intelligence System | *Python, FastAPI, LLM (Claude API), SQLite, Chart.js, argparse*

- Engineered a full-stack conversational AI system that lets users query a banking data warehouse in plain English — the LLM brain translates intent into optimized SQLite SQL using runtime schema injection, few-shot prompting, and 6-turn conversation memory for follow-up context.
- Constructed a banking data warehouse (8 branches, 80 accounts, 60 loans, 300 transactions, 96 monthly KPI records) with complex multi-table joins across NPA, CASA, CD ratio, and deposit/advance metrics — all queryable via natural language.

- Designed a dark-luxury real-time chat UI (single HTML file, no build step) with auto-rendered Chart.js visualizations (bar/line/pie selected by AI), live KPI dashboard, and tabbed result views (Insight → Chart → Data → SQL).
- Implemented a dual-mode LLM agent (Anthropic Claude API + 8-intent rule-based NLP fallback) with zero-config startup — fully functional without any API key; upgrades automatically when ANTHROPIC_API_KEY is set.

Data Professional Survey Dashboard | *Power BI, MySQL, Data Cleaning*

- Analyzed survey responses from 630 data professionals to uncover insights on salary by job title, programming language preferences, job satisfaction, and geographic distribution.
- Crafted interactive Power BI visualizations including bar charts, donut charts, and gauge metrics to compare average salaries across roles (Data Scientist: $58.58K, Data Engineer: $44.77K, Data Analyst: $55.48K).
- Uncovered Python as the dominant language among data professionals and surfaced happiness scores (4.27 for salary satisfaction, 5.86 for work-life balance) to benchmark industry sentiment.
- Performed end-to-end data cleaning and transformation in MySQL to standardize raw survey data before visualization, ensuring accuracy across all dashboard metrics.

Twitter Analytics Dashboard | *Python, Power BI*

- Engineered a Power BI dashboard using Python to track tweet engagement metrics including CTR, impressions, and interactions, driving a 25% improvement in content strategy.
- Spearheaded an in-depth analysis of weekday versus weekend social media performance, identifying peak posting times and boosting strategic engagement efficiency by 20%.

EDUCATION

Poornima College of Engineering November 2021 – May 2025
B.Tech in Information Technology – CGPA: 7.26 / 10 *Jaipur, Rajasthan*

Step By Step Senior Secondary School April 2020 – May 2021
Higher Secondary (Class XII) – CBSE *Alwar, Rajasthan*

April 2018 – May 2019
Secondary (Class X) – CBSE *Alwar,Rajasthan*
(Class X) -CBSE

CERTIFICATIONS

- SQL for Data Analysis
- Generative AI
- Python